Evaluation of Egyptian Hieroglyph Classification Across Diverse Writing Styles

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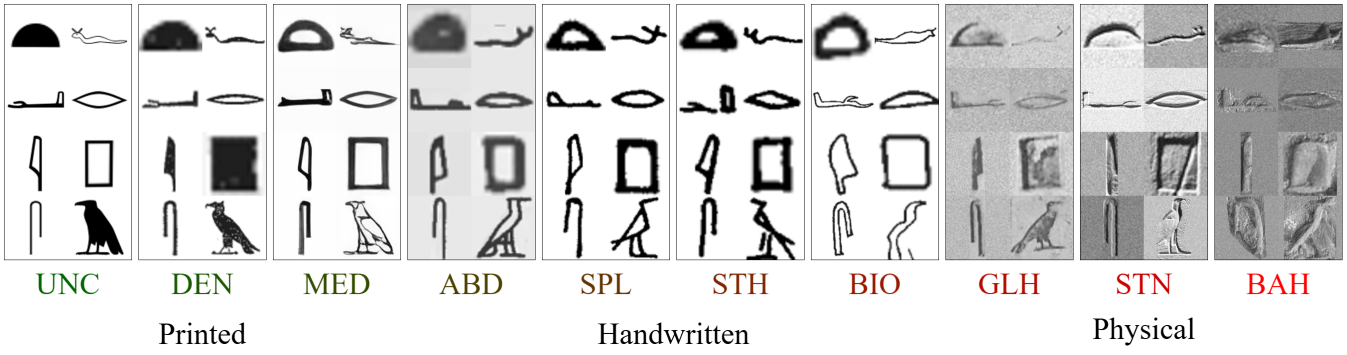


Figure 1: Structure of the MMM Dataset. We propose a multi-style dataset for Egyptian hieroglyph classification, comprising 10 groups collected from various sources, each representing a distinct writing style. The dataset is categorized into three main groups based on the creation style of hieroglyphs: printed, handwritten, and physical inscriptions. The groups are arranged in order of increasing recognition complexity, indicated by a color gradient from green to red. Our dataset serves as a starting point for solving the task of digitization of ancient Egyptian texts, providing a foundation for future research in hieroglyph recognition.

Abstract

The classification of Egyptian hieroglyphs remains a challenging problem due to the vast variability in writing styles across time periods, regions, and individual scribes. In this work, we present a comprehensive evaluation of hieroglyph classification performance across diverse stylistic domains, highlighting the limitations of current models in generalizing beyond a single style. We introduce a dataset that spans multiple writing styles, ranging from monumental inscriptions to handwritten manuscripts, and assess several near state-of-the-art recognition models. Our analysis reveals significant discrepancies in model performance when exposed to unseen styles, underscoring the need for style-aware learning strategies. This study provides a framework for future research on hieroglyph recognition with a focus on stylistic diversity and serves as a first step toward building vision-language systems capable of analyzing Egyptian hieroglyphic writings.

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CCS Concepts

• **Computing methodologies** → **Object recognition**; • **Applied computing** → *Arts and humanities.*

Keywords

dataset; hieroglyph recognition; low-resource language

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1 Introduction

The study of ancient languages has fascinated researchers for years, providing insights into the mysteries of early civilizations. These languages are like complex puzzles that, once solved, reveal details about societies that no longer exist. Researchers rely on statistical and computational methods [3, 4, 13, 31, 37] to analyze and interpret these languages, requiring text digitization. This process, often manual and error-prone, benefits from Optical Character Recognition (OCR) powered by deep learning, which accelerates digitization and reduces human effort. While deep learning has been useful for some ancient languages [33], Egyptian hieroglyphs remain underexplored.

CNNs have shown promising results for hieroglyph recognition [8, 18, 19], but most studies rely on single-source or synthetic datasets, limiting generalization. Neural networks struggle when applied to out-of-domain data [9], a critical issue for hieroglyphs, whose appearance varied over millennia across different surfaces and techniques. Training on a single-source dataset is insufficient, as demonstrated in [7], emphasizing the need for diverse, multi-source datasets to improve model robustness.

Recently, multimodal models that simultaneously process visual and textual information have emerged, enabling promising zero-shot classification capabilities [27, 35, 39]. In a preliminary evaluation of the leading models in this domain, we observed that they struggle with the intricacies of hieroglyph recognition, as illustrated in Table 1. Despite these challenges, such models hold significant potential for recognizing ancient Egyptian hieroglyphs. Unlike conventional approaches that rely on a fixed list of hieroglyphic character, a multimodal model that can visually analyze and semantically interpret hieroglyphs may offer deeper insights into their meanings. This suggests that while the current performance of these models is suboptimal, fine-tuning them with carefully labeled data could unlock a more robust and flexible recognition system, enhancing our ability to decode the complexities of ancient Egyptian writings.

To date, the only publicly available dataset of Egyptian hieroglyphs is Glyph [14], which consists of hieroglyphs sourced from a single pyramid and, therefore, presents a homogeneous style. For research in deep learning applications on Egyptian hieroglyphs to progress, new datasets are essential. However, creating these datasets requires extensive effort in labelling and data collection, demanding significant resources in terms of time, funding, and labour — resources that are especially limited in niche research areas like this. In this paper, we aim to determine how to build a hieroglyphic dataset that generalizes effectively to a wide range of unseen styles without exhaustively collecting all possible data.

Model	UNC	DEN	ABD	STH	GLH	STN
ChatGPT 4o	22	20	24	18	20	14
LLama 4 Scout	18	4	6	2	4	0
Claude 3.7 Sonnet	16	2	4	4	4	6
Gemini 2.5 Pro	42	28	34	26	28	18

Table 1: Comparison of existing multimodal models on zero-shot hieroglyph recognition. We randomly sampled 50 hieroglyphs from every style group and asked models to identify their Gardiner code. The reported results indicate accuracy in %. While models show some knowledge of Egyptian hieroglyphs, there is an evident need to tailor them to this task in order to build a robust visual-language system for ancient Egyptian writings.

To build a roadmap for effectively creating a robust dataset for hieroglyph recognition, we pose two key questions. The first is: **Do all hieroglyphic writing styles contribute equally to a model's generalization ability, or are some more effective than others?** If we identify styles that improve generalization more efficiently, we can prioritize data collection for those styles, potentially reducing the overall volume of data needed to achieve a desired level of model performance.

The second major challenge with ancient Egyptian hieroglyphs is the sheer variety of symbols. Different texts may contain vastly different sets of hieroglyphs. Thus, even if we identify the best generalizing writing styles and collect data on them, models trained on such datasets would still struggle in practical applications, as they would lack knowledge of many hieroglyphs. A common approach to addressing data scarcity in various fields is the use of synthetic data [1, 6]. This leads to the second question we explore in this work: **Can generative models create realistic hieroglyph images that expand the set of hieroglyphs that neural networks can reliably recognize?**

A diverse dataset is essential for statistically meaningful conclusions. Since collecting data for classification is easier and faster than for tasks like detection or segmentation, we focus on hieroglyph classification. However, we strongly believe that the insights gained from classification will still apply to these more advanced tasks, which are integral to OCR.

In this work, we introduce **Mystic Multisource Monoglyphs** (**MMM**) dataset, the first multi-style dataset for hieroglyph classification. The dataset comprises 10 distinct groups, each capturing a unique writing style (see Figure 1) meticulously assembled through manual labeling. To ensure the high annotation quality, labels are derived from professional text digitizations verified by experts. By providing diverse domain coverage along with detailed analysis, proposed dataset aims to be a guiding light for future datasets, facilitating progress in hieroglyph recognition.

On top of the novel dataset, we provide experimental analyses addressing three central topics: (1) Domain Gap Evaluation, to explore whether certain hieroglyphic styles enhance model generalization

Dataset	Images	Classes	Dataset	Images	Classes
UNC	2766	513	UNC_{DG}	228	37
DEN	1129	218	DEN_{DG}	565	37
MED	1387	142	MED_{DG}	-	-
ABD	2934	240	ABD_{DG}	1778	37
SPL	1010	115	SPL_{DG}	691	37
STH	627	117	STH_{DG}	-	-
BIO	929	157	BIO_{DG}	578	37
GLH	4030	171	GLH_{DG}	2911	37
STN	854	150	STN_{DG}	552	37
BAH	360	103	BAH_{DG}	-	-
Total	16015	513	$Total_{DG}$	7075	37

Table 2: Dataset Statistics. The table displays the number of images and unique classes for each group in the full dataset and the domain gap (DG) subset. All groups except GLH are the contribution of this paper. Groups not included in the DG subset are marked with "-" in relevant fields. DG subsets contain only the classes common across all selected groups, resulting in an equal number of unique classes (37) and a reduced image count compared to the full version.

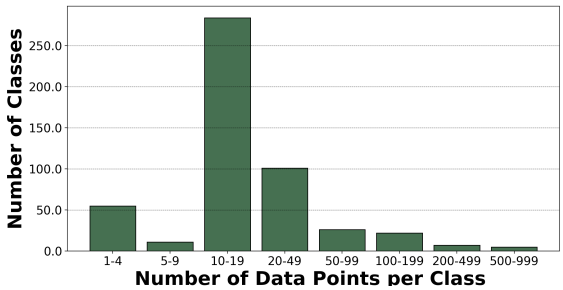


Figure 2: Distribution of Classes by Number of Data Points per Class. This histogram illustrates the long-tailed distribution of hieroglyphic classes based on the number of data points per class. The x-axis represents the range of data points for each class, and the y-axis indicates the number of classes within each range. Most classes contain a low number of samples (1-19 data points), while a few classes have significantly more, which adds complexity to the task.

more effectively; (2) Synthetic Data Evaluation, to assess the potential of generative models to produce realistic images that expand the range of recognizable hieroglyphs; and (3) an Classification Performance Analysis, to establish baseline results and practical insights for future research.

To sum up, this work makes several key contributions to the study of Egyptian Hieroglyph Recognition:

- We introduce the **MMM**, the first multi-style hieroglyph classification dataset, tripling the amount of publicly available labelled hieroglyph data and providing diverse coverage of different writing styles.
- We conduct experiments to investigate how hieroglyphic style diversity influences model generalization and evaluate the effectiveness of training with synthetic data, providing insights into efficient hieroglyph dataset construction.
- To support efficient future data collection and annotation efforts, we propose a pipeline for synthetic data generation.

2 Related works

Before neural networks, various classical machine learning approaches addressed hieroglyph recognition. In [14], the authors explored image descriptors and retrieval techniques, showing improved prediction accuracy when integrating a language model. This study introduced Glyph, the first open-source dataset in this field, which we utilize in our work. Another study [12] focused on edge detection and template matching using Histogram of Oriented Gradients descriptors.

The advent of CNNs brought notable advancements in this domain. In [8], CNNs were evaluated on Glyph and a proprietary dataset, with a lightweight architecture outperforming larger models, though fine-tuning was not explored. Another study applied Detectron2 for instance segmentation on the same dataset [18]. A combined task of recognition and translation was undertaken in [32], and [2] explored CNNs on a dataset of hieroglyphs drawn by crowd-sourced participants, though it remains proprietary. CNNs have also been used for semantic segmentation in hyperspectral imaging of Egyptian artifacts [10].

Despite these advancements, existing studies either rely solely on the Glyph dataset or use proprietary data that is not publicly available. We are the first to create, share, and explore a dataset that encompasses multiple hieroglyphic styles, providing a diverse and open resource for advancing egyptian hieroglyph recognition research.

In terms of synthetic data generation, there is limited research on ancient languages [5, 26, 29]. The study [1] focuses on generating script on Aramaic sculptures using 3D models. There is a work [15] related to hieroglyph generation. However, it involves creating a digital font rather than generating realistic hieroglyph images. Thus, we are the first to apply modern generative models for style-aware synthetic hieroglyph image generation to improve hieroglyph classification.

3 Mystic Multisource Monoglyphs Dataset

The **MMM** dataset brings together hieroglyphic material that spans the full spectrum of real-world sources from clean digital facsimiles and hand-drawn book illlustration to heavily weathered monument photographs. This breadth offers a representative sample of how hieroglyphs appear under widely differing recording conditions and writing conventions, making the collection valuable for studying style transfer and domain robustness. To minimize copyright concerns, we release only the tightly cropped images of individual hieroglyphs together with their Gardiner-code labels; the surrounding page or monument context is omitted. Because each crop is small, non-textual, and strictly transformative of the original images, we believe distribution under fair-use principles is justified while still giving researchers high-quality visual data.

3.1 Dataset construction

Data collection. While hieroglyphs exist in numerous historical sources, we focused on texts that had already been digitized and made available through the Thesaurus Linguae Aegyptiae (TLA) [28], a crucial resource for the labeling process described later. This platform provides collections of hieroglyphic texts along with translations and references to their original sources. Using these references, we located the original sources, either book scans or photographs, analyzed their writing styles, and selected the most distinctive ones. As a result, we collected 19 images from eight different sources, which were then manually labeled. Additionally, we constructed a synthetic dataset using six different Unicode fonts for Egyptian hieroglyphs, forming the **UNC** group. Finally, we incorporated the Glyph [14] to enhance the robustness of our experiments. We preprocessed it to match other data and designated this dataset as the **GLH** group. The mentioned 8 sources, along with **UNC** and **GLH**, form the 10 style groups that constitute our dataset.

To define classes in considered classification problem, we use the Gardiner's Sign List [16]. This is a widely adopted system for categorizing Egyptian hieroglyphs based on the objects they depict. Hieroglyphs are organized into groups, each marked by a letter that denotes a general category, such as animals, plants, or tools. Within each group, individual hieroglyphs are assigned a sequential number, creating a unique code (e.g., "A1" for the first hieroglyph in Group A).

The process of identifying the exact Gardiner code for a given hieroglyph image is tedious and requires significant attention and memory, as the Gardiner system comprises 763 hieroglyphs, with over 7,000 additional symbols in extended versions. Labeling was conducted by four Egyptology students and two non-specialists without a background in Egyptology. To simplify the process and reduce potential errors, we did not classify each hieroglyph individually. Instead, we transferred digitized texts from TLA onto the collected images, manually labeling the entire images using bounding boxes and polygon masks. For isolated hieroglyphs, we applied cropping with bounding boxes, while for densely packed symbols, we used polygon masks to ensure each extracted image contained only a single hieroglyph. This method allowed us to construct a hieroglyph classification dataset, where each image contains a single hieroglyph labeled with its corresponding Gardiner code. For the UNC and GLH datasets, the Gardiner codes were known in advance, so we directly assigned them without requiring additional annotation.

The labeling process was highly time-intensive, with an average annotation time of four hours per image. The fastest session took 1.5 hours, while the most complex image required 22 hours to complete. Each image contained between 117 and 2,934 hieroglyphs. Annotators reported that using TLA significantly accelerated the annotation process by providing consistent reference materials and aiding in the identification of ambiguous hieroglyphs. The labeling was conducted as part of an educational course, with no financial compensation provided.

The use of TLA digitizations played a crucial role in maintaining annotation quality, mitigating the risks associated with using non-professional or low-expertise annotators. Since TLA digitizations are verified by professional Egyptologists, there is a high level of confidence in the accuracy of the assigned labels. Additionally, if a hieroglyph in the digitized text and its corresponding image were visibly different, annotators marked the discrepancy, and such hieroglyphs were excluded from the dataset. Finally, to further enhance the dataset's reliability, a PhD-level Egyptologist reviewed all labeled data, including the **GLH** group, to identify and correct any potential errors.

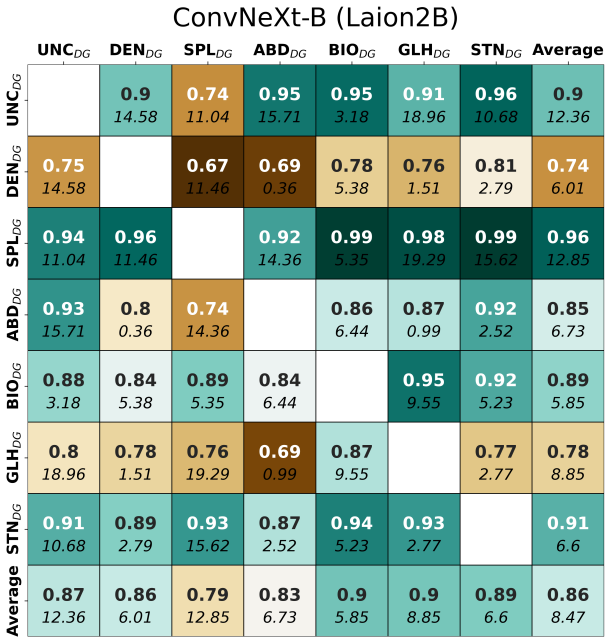


Figure 3: Domain Gap Evaluation. This heatmap visualizes domain gaps between groups by comparing model performance and data distributional distances. The upper values represent micro-F1 scores (↑) for the best generalizing model in our experiments, ConvNeXt-B (Laion2B), trained on each group (row) and evaluated on others (column). The lower values denote the FID (↓) between datasets, which measures the distributional difference between two groups independently of the model. The results show that some datasets generalize better than others, while FID and F1 scores are not strongly correlated.

3.2 Dataset statistics

The dataset statistics are presented in Table 2. Each group varies in both the number of images and the number of unique classes, highlighting the diversity within this dataset. The **UNC** group contains instances from all classes present in the dataset, making its class count equal to the total number of unique classes. Additionally, the histogram in Figure 2 illustrates the distribution of samples per class, showing a long-tailed distribution. This imbalance adds complexity to the task, as models must generalize across classes with varying representation levels.

We do not provide a predefined train/test split, as we focus on evaluating performance on unseen data, which aligns with the aim

Model	Training Datasets							Average
	$\mathbf{UNC}_{DG}$	$\mathbf{DEN}_{DG}$	$\mathbf{SPL}_{DG}$	$\mathbf{ABD}_{DG}$	$\mathbf{BIO}_{DG}$	$\mathbf{GLH}_{DG}$	$\mathbf{STN}_{DG}$	
ConvNeXt-T (IN1k)	0.843	0.639	0.908	0.783	0.800	0.723	0.778	0.782
ConvNeXt-T (IN12k)	0.876	0.684	0.927	0.780	0.828	0.762	0.836	0.813
ConvNeXt-T (IN22k)	0.877	0.654	0.919	0.789	0.839	0.758	0.799	0.805
ConvNeXt-S (IN1k)	0.848	0.620	0.915	0.770	0.830	0.735	0.802	0.789
ConvNeXt-S (IN12k)	0.889	0.678	0.926	0.802	0.845	0.766	0.855	0.823
ConvNeXt-S (IN22k)	0.900	0.672	0.935	0.837	0.860	0.788	0.851	0.835
ConvNeXt-B (IN1k)	0.871	0.653	0.895	0.805	0.820	0.762	0.816	0.803
ConvNeXt-B (IN22k)	0.882	0.696	0.919	0.797	0.835	0.790	0.851	0.824
ConvNeXt-B (Laion2B)	**0.901**	**0.742**	**0.962**	**0.854**	**0.887**	0.780	**0.911**	**0.862**
EfficientNet-B0 (IN1k)	0.589	0.482	0.685	0.600	0.459	0.399	0.541	0.537
EfficientNet-B1 (IN1k)	0.629	0.462	0.720	0.641	0.369	0.356	0.538	0.531
EfficientNet-B2 (IN1k)	0.656	0.513	0.781	0.675	0.455	0.381	0.580	0.577
ResNet34 (IN1k)	0.459	0.331	0.488	0.395	0.371	0.414	0.238	0.385
ResNet50 (IN1k)	0.382	0.334	0.476	0.380	0.370	0.403	0.274	0.374
ResNet101 (IN1k)	0.389	0.298	0.483	0.377	0.311	0.392	0.242	0.356
SwinV2-T (IN1k)	0.820	0.609	0.907	0.763	0.836	0.746	0.734	0.773
SwinV2-S (IN1k)	0.842	0.628	0.921	0.777	0.834	0.732	0.781	0.788
SwinV2-B (IN22k)	0.880	0.678	0.933	0.801	0.876	0.780	0.863	0.830
SwinV2-L (IN22k)	0.894	0.703	0.947	0.836	0.881	**0.797**	0.883	0.849
Average	0.759	0.583	0.824	0.709	0.700	0.645	0.693	0.702

Table 3: F1 Scores for Domain Gap Evaluation. The columns represent training groups, with values indicating the average F1 score (↑) across all evaluation datasets. These scores highlight how effectively each group, when used for training individually, enables models to generalize to others. ConvNeXt-B (Laion2B) and SwinV2 (IN22k) achieve the highest scores, suggesting that larger models pretrained on extensive datasets outperform others. The last row, showing the average scores across all models, suggests that SPL and UNC tend to support better generalization overall.

of this work — developing an efficient approach for collecting hieroglyphic data, potentially with the assistance of neural networks. Therefore, we treat each group as either a training or testing set in different experiments, ensuring that models are always evaluated on styles whose images have not been manually labeled for training.

4 Domain Gap Evaluation

Setting. We begin our analysis of the proposed dataset by evaluating the domain gaps between style groups. Our goal is to determine the similarity between the collected writing styles. We train models on a single group and evaluate them on all others. This setup helps identify datasets that may be particularly beneficial for training models with strong generalization capabilities. Experiments are conducted across all possible train-evaluation dataset splits. Evaluation is performed after each training epoch, and for each evaluation dataset, we report the highest achieved metric score. Complementarily, we calculate the Fréchet Inception Distance (FID) [21] between each pair of groups to measure the distributional differences between their images. This provides additional insight into the relationship between FID and classification performance on unseen data.

The proposed dataset consists of groups with distinct sets of hieroglyphic classes, making domain gap evaluation more challenging, as a model can only predict the classes it has encountered during training. To address this, we use only the following groups: **UNC**, **DEN**, **SPL**, **ABD**, **BIO**, **GLH**, and **STN**, selecting a subset of each that includes only the classes common across all these groups. These subsets are denoted with a "DG" subscript (see Table 2). The excluded groups either contain too few classes or consist primarily of unique classes, resulting in minimal overlap with other groups. Since comparing only a small number of common classes across groups could lead to misleading insights, we choose to omit them from the domain gap evaluation.

For a robust evaluation of domain gap effects, we employ a diverse set of architectures to minimize architecture-specific biases. Our baseline models are selected to achieve three objectives: analyzing domain gap effects across distinct architectural designs, across various model sizes, and across pretraining datasets of different scales.

To ensure architectural diversity, we include classical, resource-efficient, and near-SOTA models. These comprise ResNet [20], EfficientNet [34], ConvNeXt [25], and SwinV2 [24]. To account for model size variability, we include at least three sizes per architecture. For pretraining diversity, each ConvNeXt size is tested with

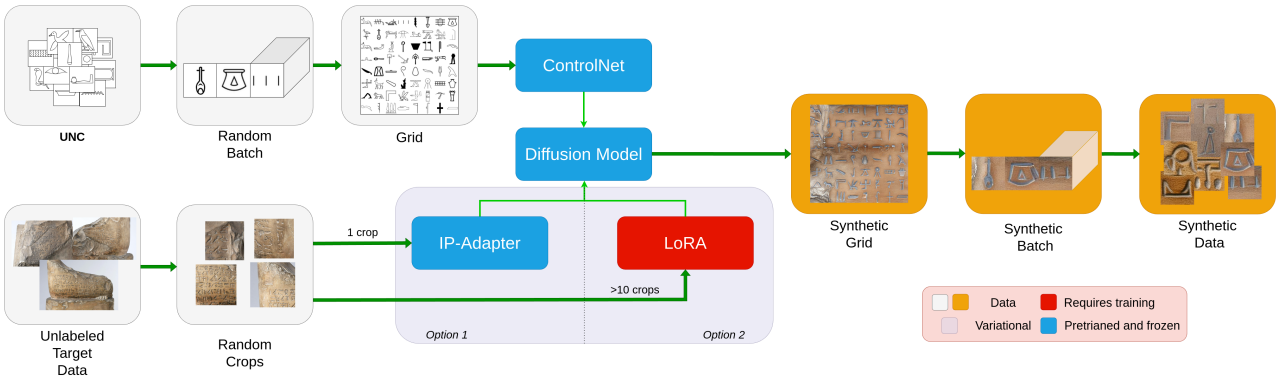


Figure 4: Synthetic data generation pipeline for Egyptian hieroglyphs. Hieroglyph class is conditioned using UNC images, passed through a pretrained diffusion model via ControlNet. Target styles are defined using random crops from style-specific images. Style adaptation is achieved with either the pratraned IP-Adapter (single style crop) or LoRA (requiring more than 10 style crops and training).

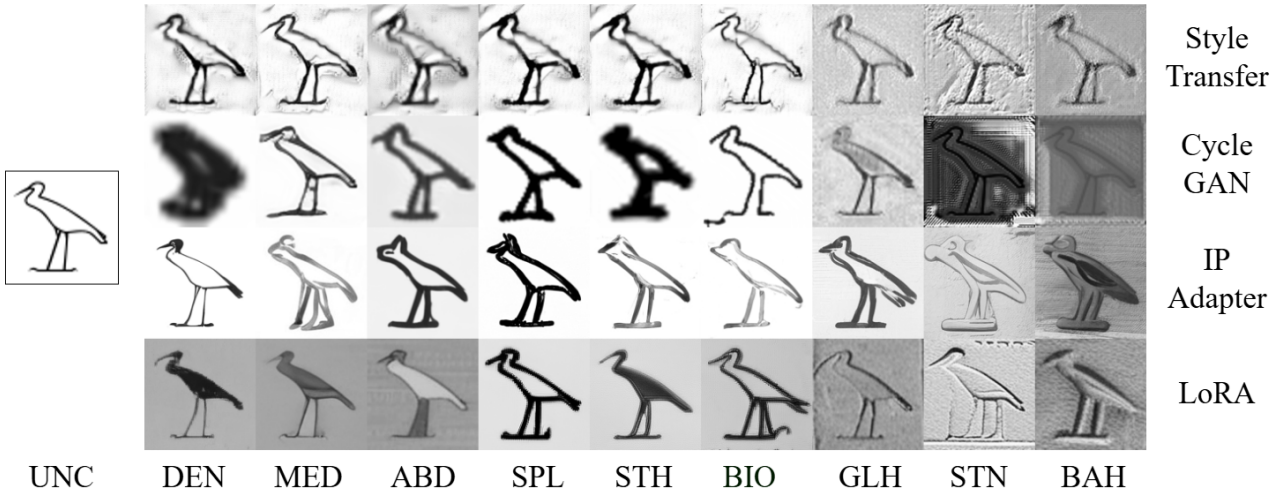


Figure 5: Synthetic Data Examples for Egyptian Hieroglyphs Across Different Styles. This figure illustrates synthetic hieroglyph images generated using four style transfer methods applied to the UNC as the base. Each row represents a different data generation method, while each column shows the group which style was simulated. Th synthetic data that matches the visual characteristics of target styles, helping to bridge the domain gap and improve model performance on less-represented classes.

three pretraining datasets of different scales. All model implementations and pretrained weights are sourced from PyTorch Image Models [36], with pretraining datasets noted in parentheses.

We fine-tune these pretrained models under consistent hyperparameters, varying only augmentation settings and learning rate. These hyperparameters are originally tuned for a separate model not included in this study. While they may not be optimal for each specific baseline, they still provide reasonable performance without major training issues. We intentionally refrain from tuning these parameters for each model individually, as our primary focus is on assessing the effect of the domain gap across different architectures rather than identifying the best-performing model.

Table 3 presents the results, revealing both expected and unexpected findings that could contribute to future advancements in this field. ConvNeXt-B pretrained on Laion2B and SwinV2-L pretrained on ImageNet22k achieve the highest average performance. This confirms the widely accepted notion that **the initial high generalizability of large models enhances their performance in low-data fine-tuning scenarios**. The **SPL** and **UNC** groups emerge as the most effective individual training datasets, consistently yielding higher F1 scores and supporting generalization to other styles. This provides an interesting insight: **certain styles produce better generalizing models than others**. Notably, the **UNC** dataset, with its simplified and highly legible figures, serves as an excellent training source for large models. However,

	DEN	MED	ABD	SPL	STH	BIO	GLH	STN	BAH
UNC	0.85 / 14.6	0.83 / 9.1	0.80 / 15.7	0.72 / 11.0	0.76 / 16.9	0.63 / 3.1	0.70 / 19.0	0.54 / 10.7	0.17 / 14.1
Style Transfer									
Gatys et al. [17]	0.83 / 7.6	0.79 / 5.6	0.81 / 7.8	0.74 / 4.3	0.74 / 6.4	0.61 / 3.0	0.74 / 7.4	0.58 / 42.9	0.25 / 1.8
CycleGAN	0.59 / **0.9**	0.81 / **0.9**	0.64 / **1.5**	0.68 / **0.6**	0.71 / **1.1**	0.66 / **1.0**	0.72 / **1.1**	0.60 / 51.6	0.24 / 6.9
LoRA	**0.90** / 4.2	0.86 / 2.8	**0.83** / 3.4	**0.79** / 2.7	0.78 / 7.9	0.67 / 2.0	**0.82** / 16.2	0.65 / 30.6	0.32 / 5.4
IP-Adapter	**0.90** / 8.4	**0.88** / 3.2	**0.83** / 6.5	0.77 / 18.0	**0.81** / 8.3	**0.69** / 1.9	0.80 / 6.9	**0.67** / **1.0**	**0.41** / **0.5**
Unsupervised Domain Adaptation									
Pseudolabels	0.61	0.75	0.61	0.65	0.53	0.49	0.52	0.42	0.06
MCD	0.87	0.82	0.80	0.74	0.77	0.61	0.75	0.63	0.27
MCC	0.82	0.79	0.79	0.68	0.77	0.60	0.72	0.58	0.24

Table 4: F1 scores and FID for Synthetic Data Evaluation using ConvNext-B. Each cell displays the F1 score (↑) and FID (↓). IP-Adapter variant of proposed style-aware hieroglyph generation pipeline achieves the highest F1 scores.

smaller models trained on **UNC** struggle, showing significantly lower performance. This highlights an important finding: **while larger models can generalize effectively from clear, simple data, smaller models do not benefit in the same way**.

The FID values and more detailed results for the best-performing model are presented in Figure 3. These results reveal that **FID is not a reliable predictor of classification quality on unseen data for this task**.

5 Synthetic Data Evaluation

Synthetic data generation is crucial for Egyptian hieroglyph recognition, given that there are over 700 symbols in the basic Gardiner's Sign List, while our dataset includes only 513 symbols that appear in at least one group. Finding the remaining symbols becomes increasingly difficult due to their rarity. Synthetic data can help address this gap by generating samples for underrepresented hieroglyphs. We consider the **UNC** group as the baseline synthetic data, as it covers all hieroglyphic classes. However, as demonstrated by the Domain Gap Evaluation experiments, training exclusively on **UNC** does not yield high-quality results, highlighting the need for improved synthetic data training techniques. Since manual labeling is too time-consuming, we focus on unsupervised methods and do not consider semi-supervised approaches.

To enhance training with synthetic data, we explore two categories of methods: unsupervised domain adaptation and style transfer techniques. For unsupervised domain adaptation, we evaluate the naive pseudo-labeling approach, Minimum Class Confusion (MCC) [23], and Maximum Classifier Discrepancy (MCD) [30]. For style transfer, we test the original style transfer method from [17], CycleGAN [40], and two variations of our pipeline that incorporate either LoRA [22] or IP-Adapter [38]. The examples of synthetic data generated with those approaches are presented in Figure 5.

5.1 Style-aware Hieroglyph Generation

We propose a hieroglyph generation pipeline (see Figure 4) using SDXL, a diffusion model chosen for its broad domain support and extensive pretraining, enabling hieroglyph generation without fine-tuning. The pipeline employs two adapters: ControlNet, which ensures structural accuracy by guiding the layout with **UNC** images, and a style adapter, either IP-Adapter, requiring a single style crop, or LoRA, which requires training on multiple style images. Reference images from the target dataset define the style, allowing the model to generate hieroglyphs with realistic appearance and structure.

Since SDXL is optimized for 1024×1024 resolution but generating individual hieroglyphs at this scale is computationally expensive, we employ batch processing. Hieroglyphs are resized and arranged into a grid before generation, then split back into individual images, improving efficiency without compromising quality.

To the best of our knowledge, we are the first to propose utilizing this setup of diffusion models and combination of two adapters for style-aware hieroglyph generation. However, as the focus of this work is the dataset, we do not conduct an in-depth analysis of this pipeline or provide detailed implementation specifics. Instead, we present a proof of concept demonstrating that this approach can produce plausible results, potentially achieving state-of-the-art performance.

5.2 Experiments

To evaluate the considered methods, we trained classification models for each group using **UNC** as labeled data and the group's data as unlabeled data, then assessed model performance on the same group's labeled data using the F1-score. For unsupervised domain adaptation, UNC was the source domain and unlabeled group data the target. For style transfer, we generated synthetic data by applying style images to UNC images (see Figure 5) and trained models on both real and synthetic data. FID measured generative quality. Since full groups were used, results are not directly comparable to domain gap evaluation metrics.

When selecting classification models, we chose the best performing architectures from the previous experiment: ConvNeXt-B

Train	DEN	MED	ABD	SPL	STH	BIO	GLH	STN	BAH
Real Data	70.4	83.2	81.3	85.3	86.0	70.1	82.9	76.7	**59.3**
+ UNC	82.6	88.4	85.6	87.9	91.0	75.2	83.1	78.5	52.8
+ Synt (Eval)	**86.8**	90.3	**89.4**	87.5	92.6	**75.6**	85.3	81.3	56.1
+ Synt (All)	84.8	**90.7**	**89.4**	**88.0**	**94.0**	74.0	**85.8**	**82.7**	51.2

Table 5: Classification Performance Analysis for ConvNeXt-B. The reported values are micro-F1 score (↑). Adding UNC improves generalization, while improved synthetic data further enhances performance, though its effectiveness varies across datasets.

(Laion2B) and SwinV2 (IN22k). Additionally, we included EfficientNet-B1 (IN1k), an average-performing model, to ensure that the results are not only valid for the best generalizing models.

The results in Table 4 clearly demonstrate that **training on synthetic data can significantly enhance model performance**. While [17] and CycleGAN provide unstable improvements and may even degrade performance, the proposed pipeline consistently achieves stable and significant improvements for both IP-Adapter and LoRA variants. Regarding domain adaptation methods, pseudolabeling severely worsens results, while MCC remains unstable. In contrast, MCD can notably improve results in some cases but still lacks consistency and does not perform as well as the proposed approach.

An interesting finding is that while CycleGAN achieves the best FID score, and the LoRA variant of our pipeline produces the most visually plausible results (see Figure 5), neural networks find synthetic data from the IP-Adapter variant to be more useful for training, underscoring the importance of experimental validation when assessing the practical usefulness of synthetic data

6 Classification Performance Analysis

We used the obtained insights to evaluate the best possible generalization performance achievable when training on our full dataset. In this experiment, we did not treat **UNC** as a separate group but rather as synthetic data. For each group, we conducted four data ablation experiments. First, models were trained on all groups except the considered one. Then, **UNC** data was added to the training set. Next, we incorporated synthetic data generated with our pipeline for the considered group. Finally, we added synthetic data for all groups.

For classification models, we used ConvNeXt-B (Laion2B) and SwinV2 (IN22k) as the best-performing architectures from previous experiments. Additionally, we included ConvNeXt-XL and Davit-H [11], two SOTA models, to examine whether increasing model size further improves performance.

The experiments showed that ConvNeXt-B (Laion2B) remains the best-performing model, indicating that the dataset size is still insufficient for larger models to reach their full potential. As seen in Table 5, adding **UNC** improves generalization by introducing unseen classes, with further gains from improved synthetic data. For **BAH**, synthetic data worsens performance, highlighting the need for more advanced hieroglyph generation methods.. Despite these improvements, the achieved performance remains far from the scores required for reliable OCR (0.95 and more) and full text digitization, highlighting the critical need for further dataset expansion.

7 Dataset Extension Roadmap

Based on our findings, we propose a two-pronged approach to extend the dataset. First, for writing styles identified as better generalizers, additional data collection should focus on expanding the set of unique hieroglyphs. Our experiments suggest that introducing new hieroglyph classes in these styles could improve their classification quality across other styles as well. Second, for styles where even well-generalizing models struggle, additional labeling and fine-tuning are essential to enhance recognition performance.

To support this effort, we adopt a human-in-the-loop framework, where deep learning models generate initial predictions and annotators review or correct them. This approach improves annotation efficiency when model outputs are accurate and enables focused synthetic data generation when model predictions fail, enhancing both coverage and quality.

Looking ahead, we plan to extend this dataset along two directions. The first targets full-scene OCR with richer layout and structural annotations. The second focuses on multimodal learning by linking hieroglyph images with transliteration and translation. Advancing both fronts is crucial for building intelligent tools that support Egyptologists in reading, interpreting, and organizing ancient texts, moving toward the long-term goal of a practical research assistant for the field.

8 Conclusion

In this work, we have addressed the challenging task of Egyptian hieroglyph classification by introducing the first multi-style dataset **MMM**, which significantly expands the scope of available hieroglyphic data. Our dataset, comprising 10 distinct style groups, provides a rich and diverse resource that captures the wide variability in hieroglyphic writing. Our analyses reveal two important findings. First, not all hieroglyphic writing styles contribute equally to model generalization, suggesting that selective data collection and prioritization of effective styles may reduce the overall annotation burden while improving performance. Second, our investigation into synthetic data highlights its potential to augment the available training material, thereby broadening the repertoire of hieroglyphs that can be reliably recognized.

Future work will focus on expanding the dataset by following our proposed dataset extension roadmap, which combines targeted data collection, synthetic data generation, and a human-in-the-loop annotation. This extended dataset would lay the groundwork for leveraging large multimodal models to support and accelerate Egyptological research, enabling powerful systems for the analysis, translation, and interpretation of ancient texts.

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